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COURSE CODE: SBT-DF202

Lab1: DIGITAL FORENSICS CASE HANDLING, AUTOPSY AND SLEUTH KIT ANALYSIS

DATE: 28/08/26

INSTRUCTOR: MR AMINU IDRIS

Download the authorised training forensic image **Ch01InChap01.dd** and create a dedicated forensic working directory before beginning your analysis. If the direct download does not start automatically, use the **browser download page** instead. The source lab specifically provides this evidence image for the practical.

After downloading the evidence image, record the evidence source, file size, storage location, MD5 hash and SHA-256 hash before conducting any forensic analysis. Preserve the original image and perform recovery and examination activities in separate working folders.

Part A — Autopsy Analysis

Create a forensic case in Autopsy and add **Ch01InChap01.dd** as the evidence source. Identify deleted files, recover relevant files, perform keyword searches, tag relevant evidence and generate an Autopsy report. Record the case name, evidence source, deleted-file findings, recovered files, keyword-search results and reporting evidence.

ANSWER:

FILE NAME: Ch01InChap01.dd

FILE SIZE: 1,440 KB

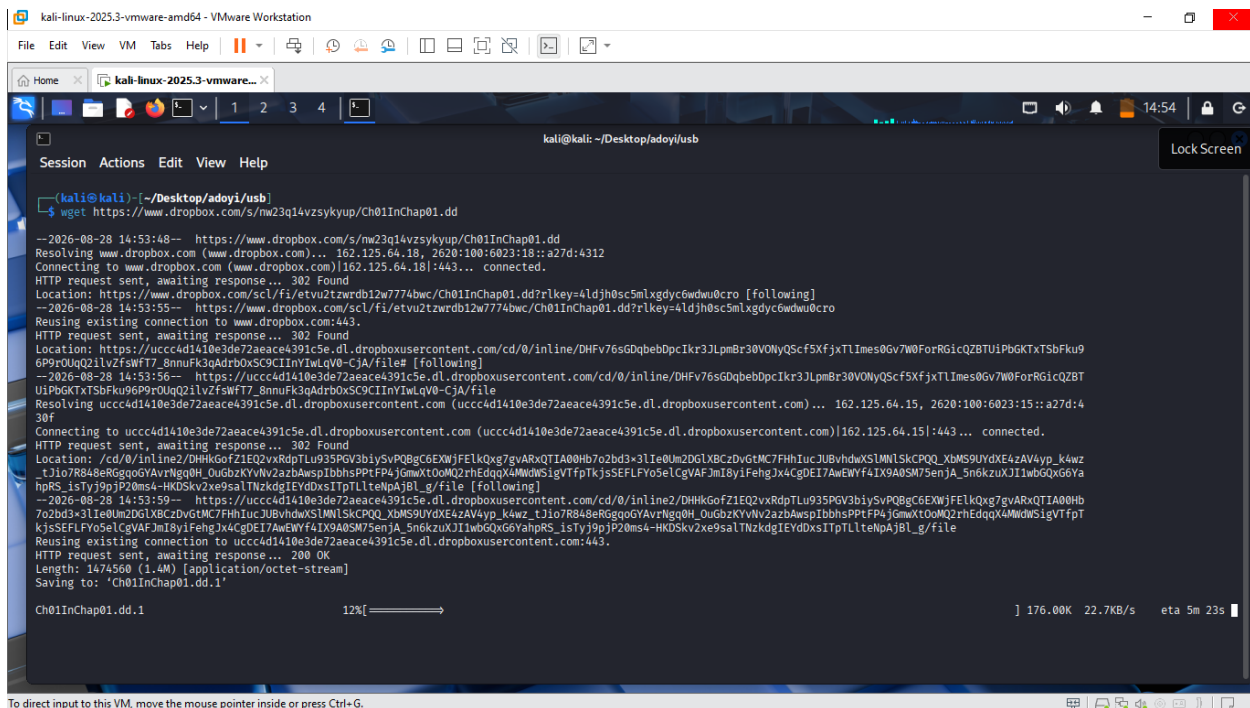
STORAGE/LOCATION: **HomePc/Desktop/lcdfa/Forensic investigation/Ch01InChap01.dd**

MD5 hash: **a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd**

SHA-256 hash: **Image Type: raw**

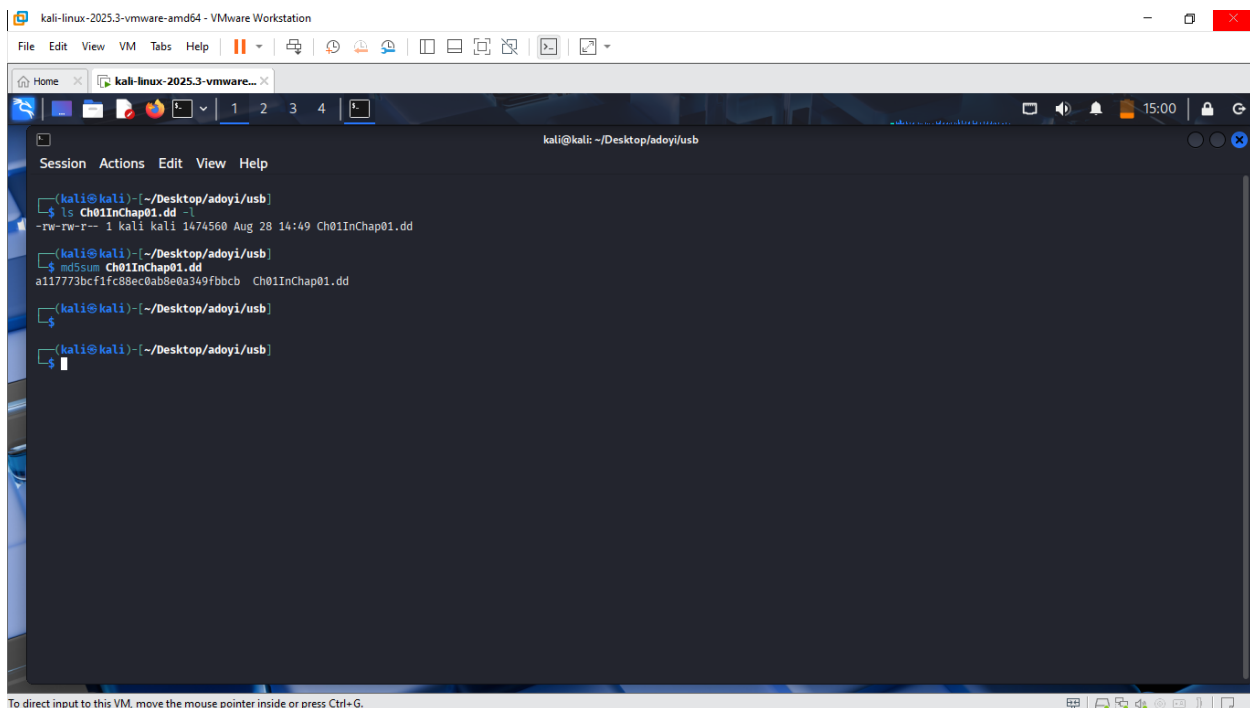
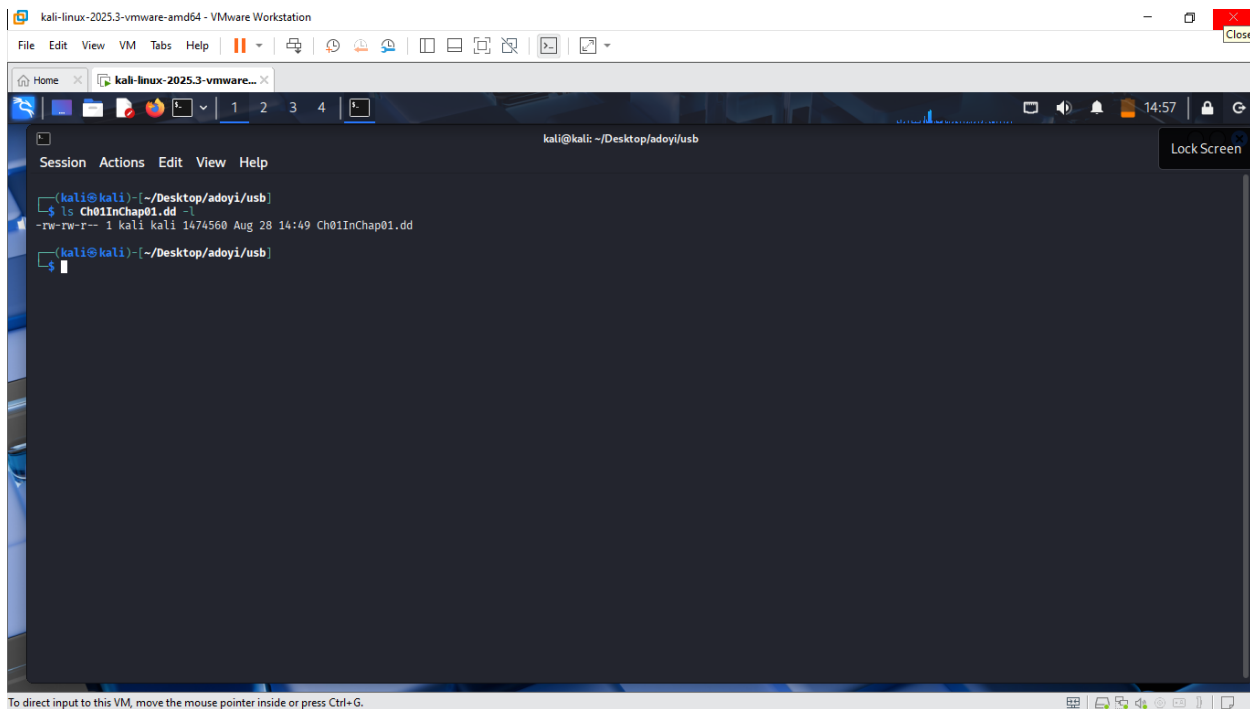
Size in bytes: **1474560**

Sector size: **512**

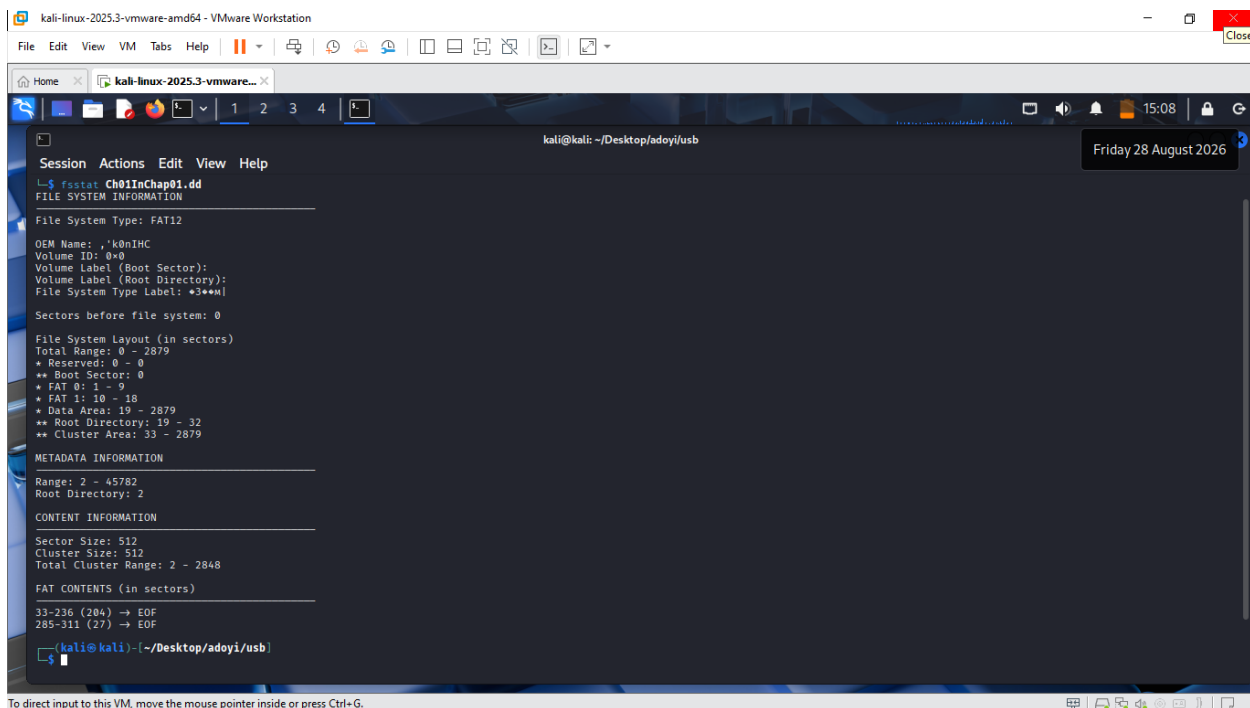
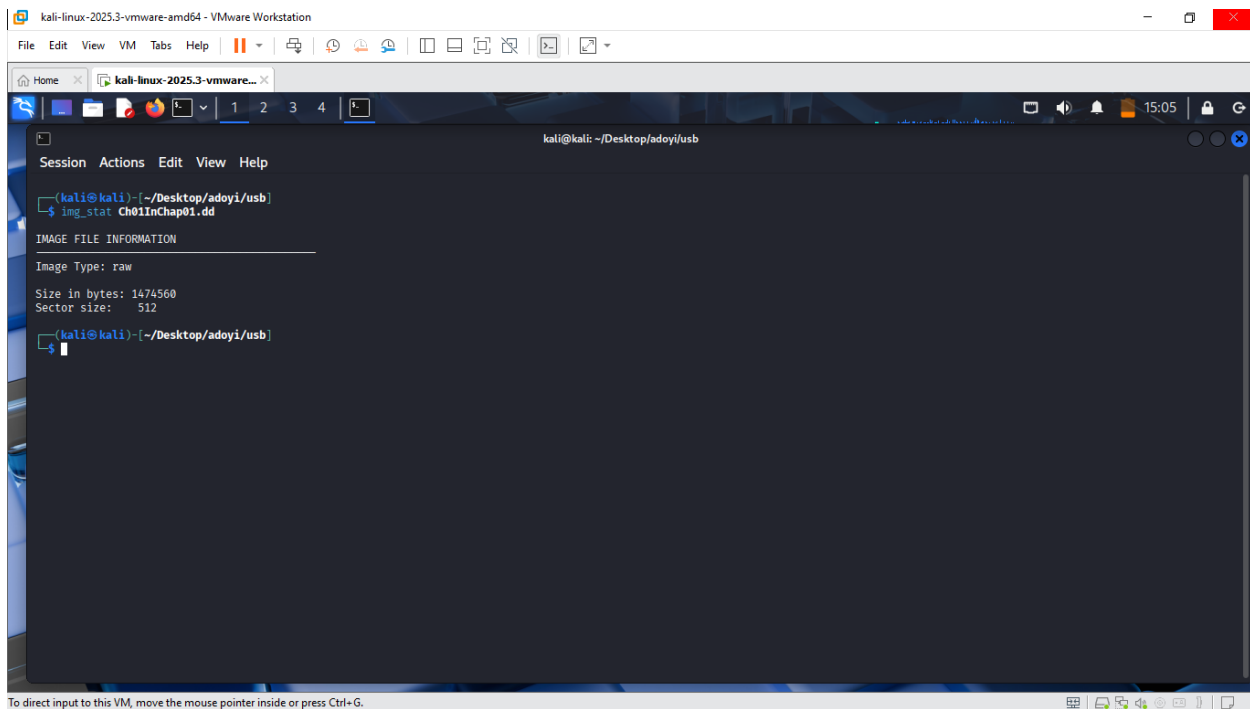


```
kali@kali: ~/Desktop/adoyi/usb
$ wget https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
--2026-08-28 14:53:48-- https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd
Resolving www.dropbox.com (www.dropbox.com)... 162.125.64.18, 2620:100:6023:18::a27d:4312
Connecting to www.dropbox.com (www.dropbox.com)|162.125.64.18|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://www.dropbox.com/sc/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mLxgdydc6wdwu0cro [following]
--2026-08-28 14:53:55-- https://www.dropbox.com/sc/fi/etvu2tzwrdb12w7774bwc/Ch01InChap01.dd?rlkey=4ldjh0sc5mLxgdydc6wdwu0cro
Reusing existing connection to www.dropbox.com:443.
HTTP request sent, awaiting response... 302 Found
Location: https://ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com/cd/0/inline/DHFv76sGDqbeDpcIKr3JLpm8r30V0NyQScf5XfjxTLImes0Gv7W0ForRGicQZBTU1PbGKTXtSbFku9
6P9r0Uq2l1vZfswF77_8nnuFk3qAdrb0xSC9CIInYwLqV0-CJA/file# [following]
--2026-08-28 14:53:56-- https://ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com/cd/0/inline/DHFv76sGDqbeDpcIKr3JLpm8r30V0NyQScf5XfjxTLImes0Gv7W0ForRGicQZBT
U1PbGKTXtSbFku96P9r0Uq2l1vZfswF77_8nnuFk3qAdrb0xSC9CIInYwLqV0-CJA/file
Resolving ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com (ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com)... 162.125.64.15, 2620:100:6023:15::a27d:4
30f
Connecting to ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com (ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com)|162.125.64.15|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: /cd/0/inline2/DHhKGoFz1EQ2vxRdpTLu935PGV3biySvPQBgC6EXWjFELkQxg7gvaRqQTIa00Hb7o2bd3x3lIe0Um2DGLXBCaDyGtMC7FhhIucJUBvhdwXSLWNLSkCPQQ_XbMS9UYdXE4zAV4yp_k4wz
_tJio7R848eRGqoGAvrNgq0H_OuGbZKYVw2azbAwspIbbhsPPTFP4jGmmXtOoMQ2rhEdoqX4MmDWSigVTFpTkjsSEFLFY05eLcgVAFJmI8y1FehgJx4cGDEI7AwEWYf4IX9A0SM75enJA_5n6kzuXJI1wbGQxG6Ya
hprS_istYj9pP20ms4-HKDSkv2xe9saITNzkdGIEYDxsITpTLtleNpAJ8l_g/file [following]
--2026-08-28 14:53:59-- https://ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com/cd/0/inline2/DHhKGoFz1EQ2vxRdpTLu935PGV3biySvPQBgC6EXWjFELkQxg7gvaRqQTIa00Hb
7o2bd3x3lIe0Um2DGLXBCaDyGtMC7FhhIucJUBvhdwXSLWNLSkCPQQ_XbMS9UYdXE4zAV4yp_k4wz_tJio7R848eRGqoGAvrNgq0H_OuGbZKYVw2azbAwspIbbhsPPTFP4jGmmXtOoMQ2rhEdoqX4MmDWSigVTFpT
kjsSEFLFY05eLcgVAFJmI8y1FehgJx4cGDEI7AwEWYf4IX9A0SM75enJA_5n6kzuXJI1wbGQxG6YahpRS_istYj9pP20ms4-HKDSkv2xe9saITNzkdGIEYDxsITpTLtleNpAJ8l_g/file
Reusing existing connection to ucccd4d140e3de72aace4391c5e.dl.dropboxusercontent.com:443.
HTTP request sent, awaiting response... 200 OK
Length: 1474560 (1.4M) [application/octet-stream]
Saving to: 'Ch01InChap01.dd.1'

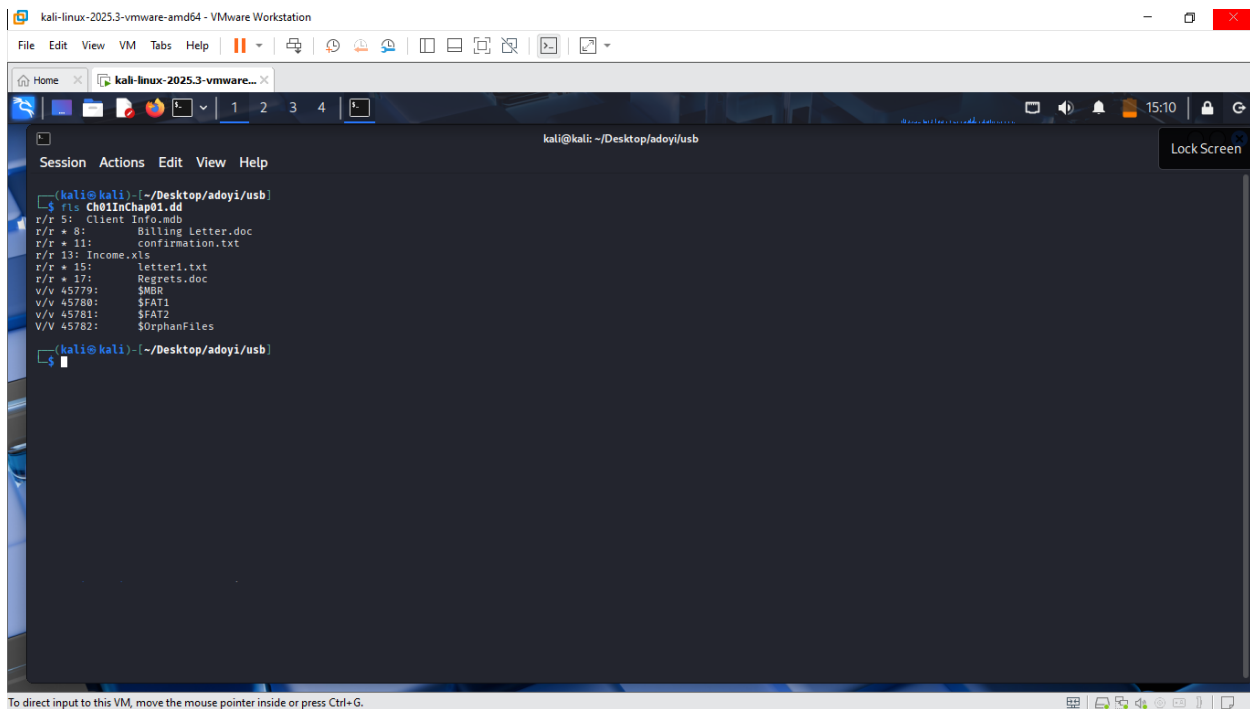
Ch01InChap01.dd.1          12%[=====>] 176.00K  22.7KB/s  eta 5m 23s
```



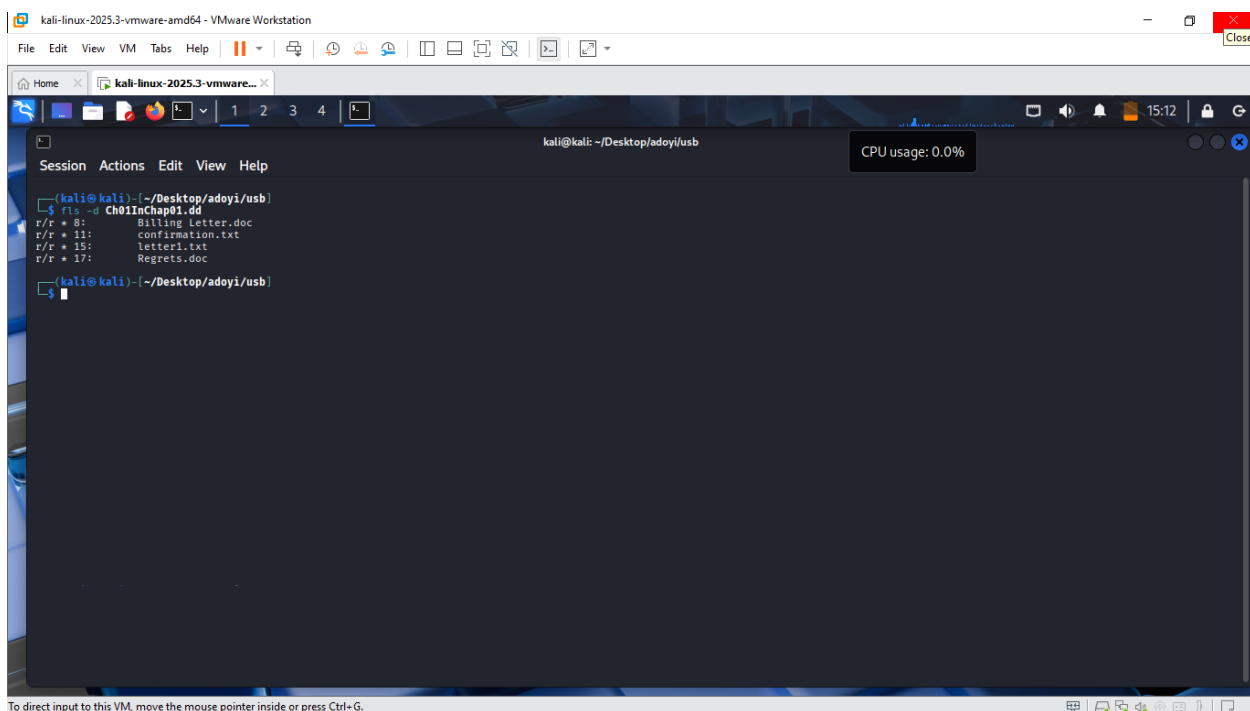
FILE DETAIL BELOW:



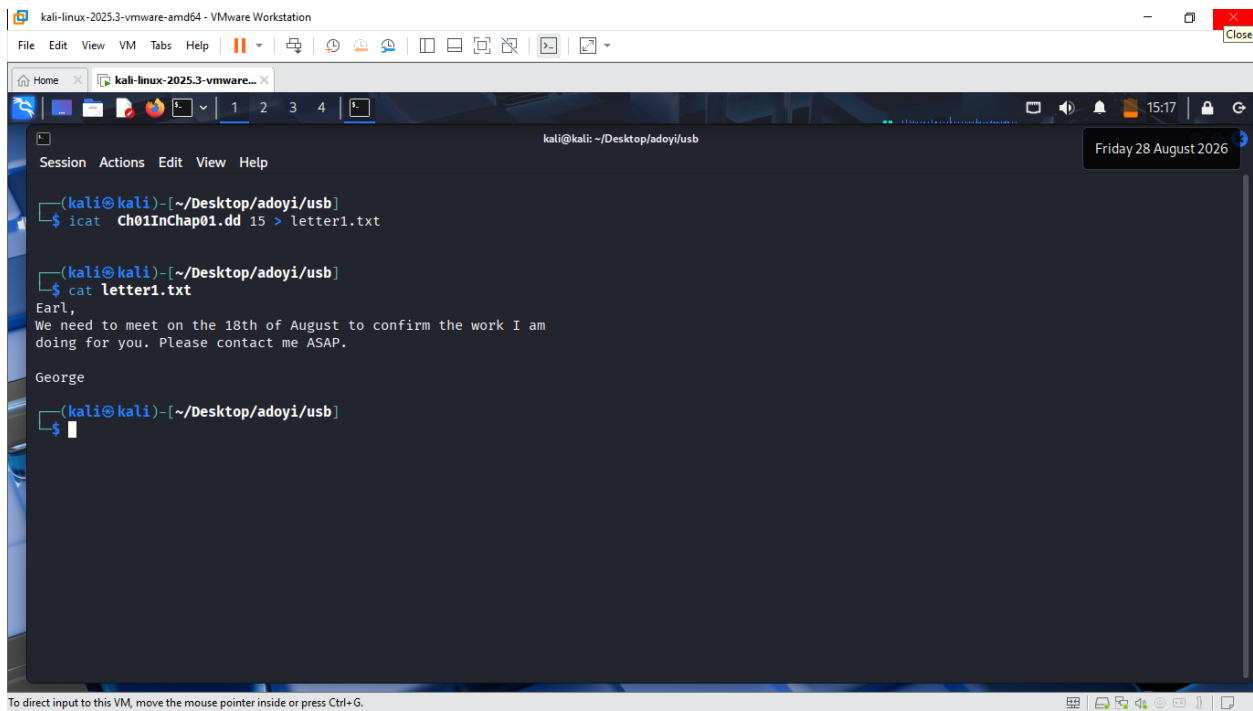
IDENTIFICATION OF DELETED FILES: ANSWER



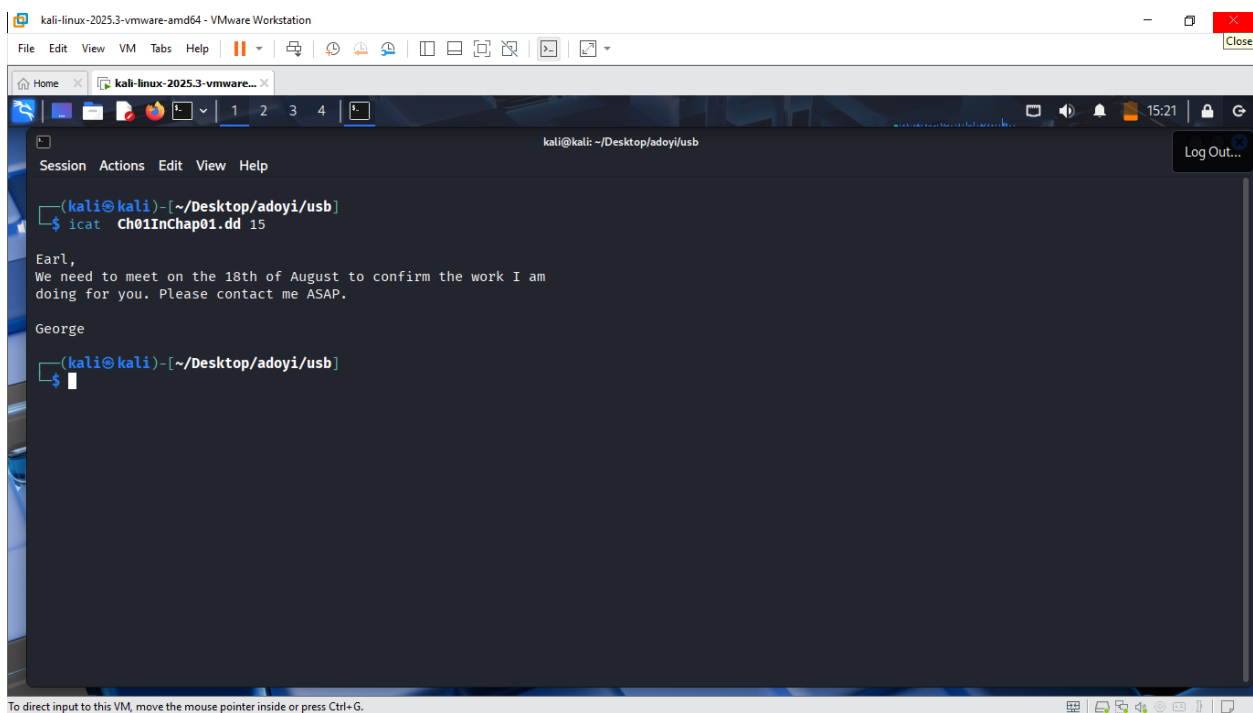
DISPLAYING OF DELETED FROM THE DIRECTORY:



RECOVERING OF DELETED FILES USING INODE:



EXTRACTION OF THE DATA UNIT:



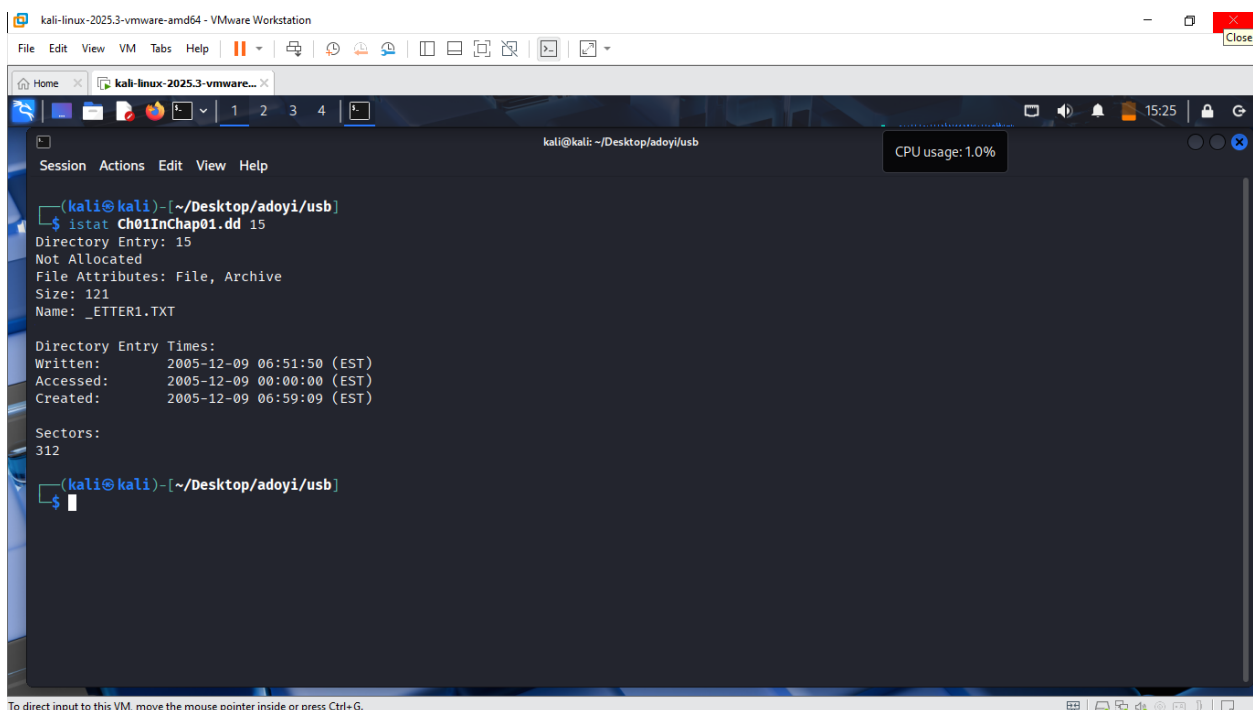
Part B — Sleuth Kit Analysis

Examine the forensic image using `img_stat`, `mmls`, `fsstat` and `fls`. Identify deleted files and locate **INCOME.XLS**. Determine its metadata address, examine its metadata and data units using `istat`, recover it using `icat` and `blkcat` where applicable, perform additional recovery using `tsk_recover`, and compare the hashes of the recovered copies.

Document every significant command, output, observation, error and corrective action. Screenshots must clearly support the work claimed in your report.

Do not modify the original forensic image. All recovered evidence, screenshots, command outputs and reports must be stored separately from the source image.

DISPLAYING THE STATISTIC DETAILS OF THE GIVEN METADATA:



The screenshot shows a Kali Linux terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation". The terminal is running the command `istat Ch01InChap01.dd 15`. The output displays metadata for a directory entry at sector 15, including file attributes, size, name, and timestamps.

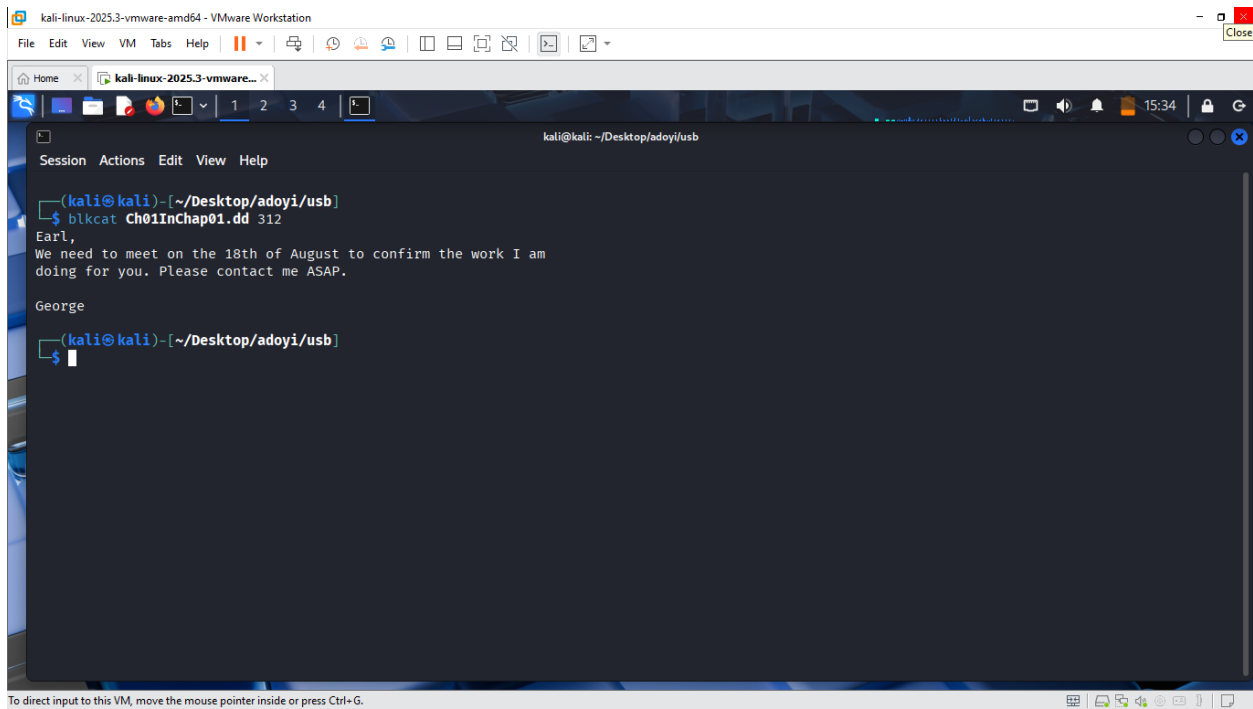
```
(kali@kali)-[~/Desktop/adoyi/usb]
$ istat Ch01InChap01.dd 15
Directory Entry: 15
Not Allocated
File Attributes: File, Archive
Size: 121
Name: _ETTER1.TXT

Directory Entry Times:
Written:      2005-12-09 06:51:50 (EST)
Accessed:    2005-12-09 00:00:00 (EST)
Created:     2005-12-09 06:59:09 (EST)

Sectors:
312

(kali@kali)-[~/Desktop/adoyi/usb]
$
```

EXTRACT THE CONTENT OF GIVRN DATA UNIT:



kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Home kali-linux-2025.3-vmware... 1 2 3 4

kali@kali: ~/Desktop/adoyi/usb

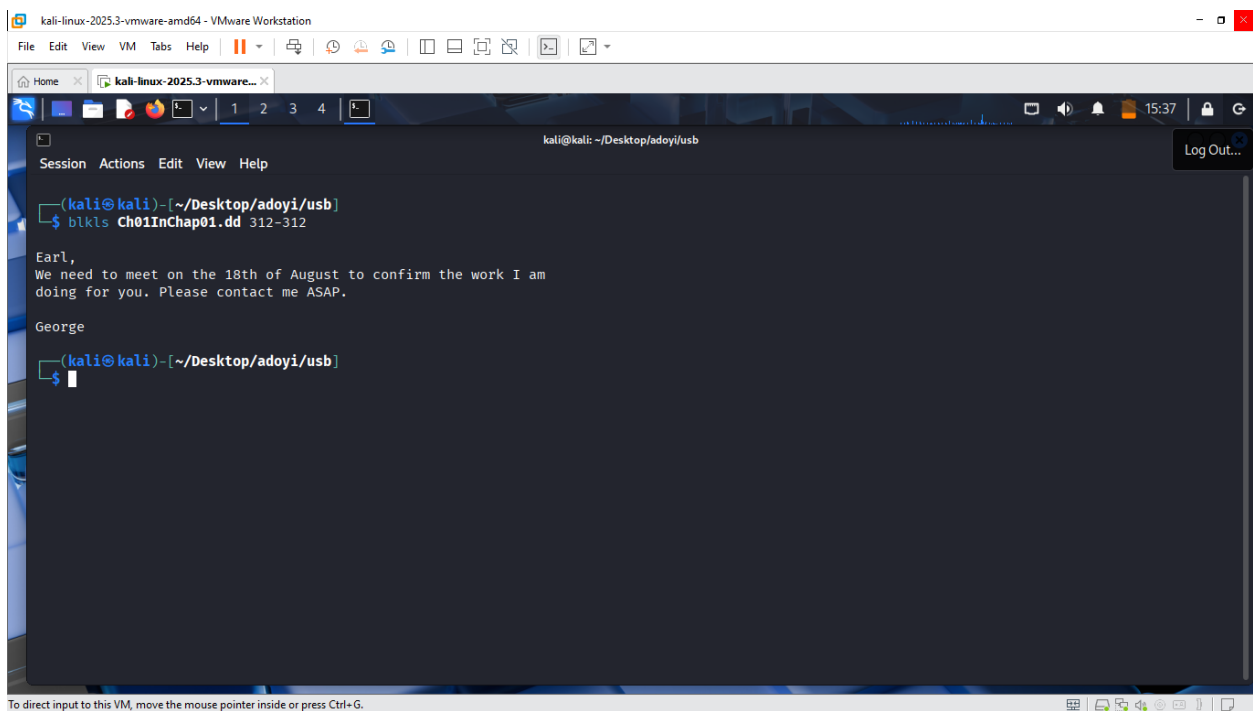
Session Actions Edit View Help

```
(kali@kali)-[~/Desktop/adoyi/usb]
$ blkcat Ch01InChap01.dd 312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
(kali@kali)-[~/Desktop/adoyi/usb]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

LIST THE DETAIL OF DATA UNIT UNALLOCATED:



kali-linux-2025.3-vmware-amd64 - VMware Workstation

File Edit View VM Tabs Help

Home kali-linux-2025.3-vmware... 1 2 3 4

kali@kali: ~/Desktop/adoyi/usb

Session Actions Edit View Help Log Out...

```
(kali@kali)-[~/Desktop/adoyi/usb]
$ blkls Ch01InChap01.dd 312-312
Earl,
We need to meet on the 18th of August to confirm the work I am
doing for you. Please contact me ASAP.

George
(kali@kali)-[~/Desktop/adoyi/usb]
$
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

EXTRACT THE CONTENT OF A GIVEN DATA IN HEX:


```
kali@kali:~/Desktop/adoyi/usb
$ bkkcat Ch01InChap01.dd 312 | xxd
00000000: 4561 726c 2c20 0d0a 5765 206e 6565 6420  Earl, ..We need
00000010: 746f 206d 6565 7420 6f6e 2074 6865 2031  to meet on the 1
00000020: 3874 6820 6f66 2041 7567 7573 7420 746f  8th of August to
00000030: 2063 6f6e 6569 726d 2074 6865 2077 6f72  confirm the war
00000040: 6b20 4920 616d 200d 0a64 6f69 6e67 2066  k t am ...doing f
00000050: 6f72 2079 6f75 2e20 506c 6561 7365 2063  or you. Please c
00000060: 6f6e 7461 6374 206d 6520 4153 4150 2e0d  ontact me ASAP..
00000070: 0a64 0a17 656f 7267 6500 0000 0000 0000  ... George.....
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001000: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001010: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001020: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001030: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001040: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001050: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001060: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001070: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001080: 0000 0000 0000 0000 0000 0000 0000 0000  .....
00001090: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000010a0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000010b0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000010c0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000010d0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000010e0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
000010f0: 0000 0000 0000 0000 0000 0000 0000 0000  .....
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

SHOWING A FILE WITH MULTIPLE SECTORS:

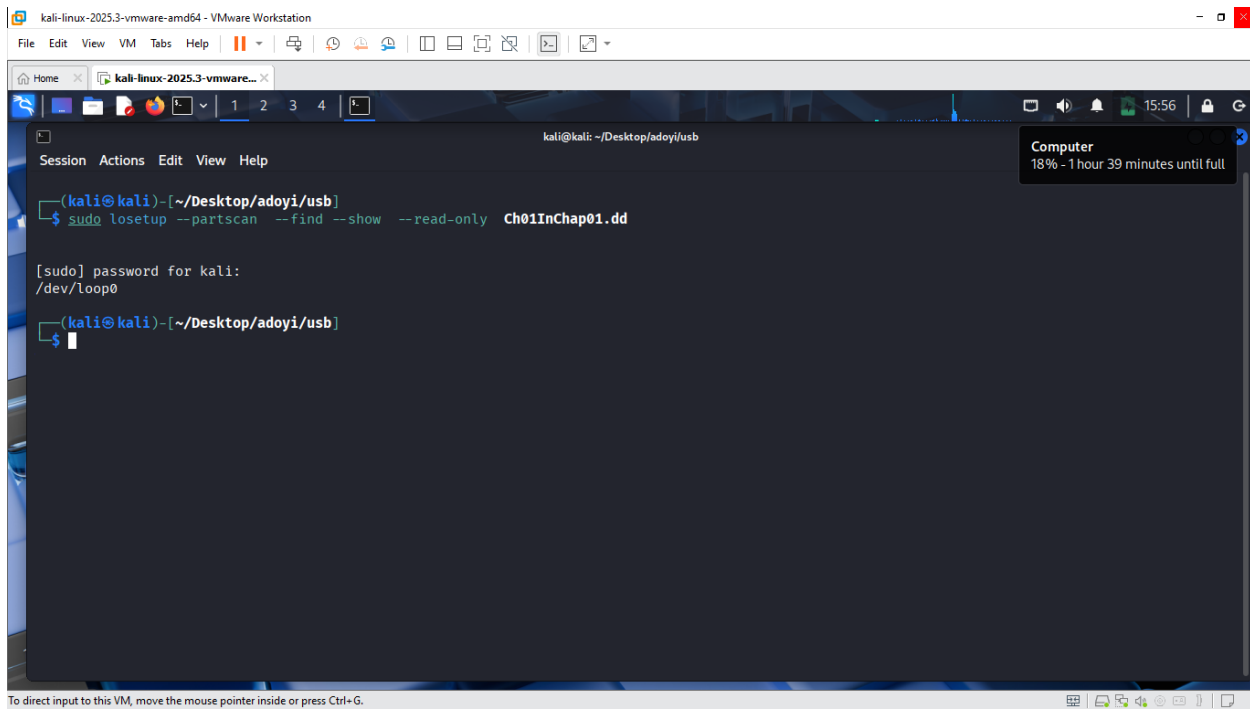
```
kali@kali:~/Desktop/adoyi/usb
$ istat Ch01InChap01.dd 13
Directory Entry: 13
Allocated
File Attributes: File, Archive
Size: 13824
Name: INCOME.XLS

Directory Entry Times:
Written: 2005-12-09 06:52:18 (EST)
Accessed: 2005-12-09 00:00:00 (EST)
Created: 2005-12-09 06:59:07 (EST)

Sectors:
285 286 287 288 289 290 291 292
293 294 295 296 297 298 299 300
301 302 303 304 305 306 307 308
309 310 311
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

MOUNTING A USB:



SHOWING THE DISK CONTENT:

